A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 36

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** Photons of light absorbed by the (n-type silicon) upper layer dislodge electrons, making it negatively charged **1** Electrons flow through an external conductor / circuit to the relatively positive lower layer – the current powers appliances **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'solar panels do not work in cold countries' - output depends on insolation, and cells are slightly more efficient when cool	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Smooth cells reflect about 30% of the light – grooved / textured surfaces (mimicking moth eyes) reflect light into the cell rather than away **1** Multi-junction cells have several layers of different materials, each absorbing different wavelengths **1** So a greater proportion of the available light is absorbed and converted to electricity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
01.3	**1 mark each** 1 mark for each of the 3 points below. **1 Solar farm:** polycrystalline / thin film - lowest cost per unit electricity where land area is not limiting (efficiency per m^2 matters less) **1 Satellite:** multi-junction - highest efficiency where mass / area is critical and cost is less important **1** Cost per watt = cost / efficiency: thin film 0.125, poly 0.125, mono 0.136, multi-junction 0.625 per unit **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
01.4	**1 mark each** 1 mark for each of the 2 points below. **1 Intermittent:** no output at night and low in winter (daily and seasonal cycles) - predictable **1 Unpredictable:** cloud cover, dust and smoke reduce intensity and scatter light and cannot be forecast accurately **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** t-test **1** The data are measured (continuous) values and two means are being compared **1** The data are (assumed to be) normally distributed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **t-test** → a test for a difference between the means of two normally distributed sets **Do not credit** • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	3

Do not write outside the box

Question	Marking guidance	Marks
02.2	1 mark for the point below. **1** Use the Mann-Whitney U test instead (compares medians) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Mann-Whitney** → a test for a difference between two sets of unpaired, non-normal data **Do not credit** • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	1
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** There is a 2% probability that the difference occurred by random chance **1** As this is less than 0.05 the difference is statistically significant / the null hypothesis is rejected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **random** → positions chosen using random numbers so the choice is not biased • **statistically significant** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **null hypothesis** → a statement that there is no significant difference or correlation **Do not credit** • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	2
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** The ranges of values (mean plus or minus SD) do not overlap: woodland up to 1.6 and clearing down to 2.4 **1** About 68% of values lie within one SD of each mean so the two data sets are clearly separated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.5	**1 mark each** 1 mark for each of the 2 points below. **1** Significance shows the difference is unlikely to be due to chance, not that a causal relationship exists **1** Another variable (eg topography / aspect of the sites) could cause both – a controlled experiment or further evidence is needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **statistically significant** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **causal** → one variable actually causing the change in the other	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Orientation:** south-facing windows (northern hemisphere) maximise passive solar gains – rooms needing warmth on the sunny side – fewer / smaller windows on the north side where losses are greatest – solar screens / brise soleil prevent summer overheating **2** Low SA:V (compact shapes, terraced / joined buildings, fewer storeys of small footprint) means less external surface per unit volume through which heat is lost **2 Earth sheltering:** the ground is warmer than winter air and shields the building from wind, so conduction and wind-chill losses are reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Cement manufacture (heating limestone) is very energy intensive so concrete has high embodied energy **1** Limecrete has about half the embodied energy of concrete – rammed earth, straw bales and lime mortar are lower still **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **embodied energy** → all the energy used to make and transport a product	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Warm stale air leaving and cold fresh air entering pass through the exchanger in opposite directions (counter-current) separated by thin conducting pipes / plates with a large surface area **1** Heat transfers from the outgoing to the incoming air so fresh air enters already warmed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-232; AQA spec 3.3.4.2.2.	
04.1	**1 mark each** 1 mark for each of the 2 points below. **1** Dissolved oxygen fluctuates daily / 24-hour cycle: highest in the afternoon (about 13-13.5 mg dm^{-3} at 16 and 40 h) and lowest at night / early morning (2.5-3.5 mg dm^{-3} at 4 and 28 h) **1** The night-time minimum is lower on the second night (2.5 vs 3.5) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 4 points below. **1** Nitrates (from fertiliser runoff) and phosphates (from sewage effluent) cause eutrophication: rapid growth of algae / cyanobacteria (algal bloom) **1** During the day photosynthesis by the dense algae releases oxygen so concentrations rise (above the normal saturation level) **1** At night there is no photosynthesis but respiration by algae / animals / bacteria continues, using oxygen **1** Aerobic bacterial decomposition of dead algae / shaded macrophytes uses oxygen – warm summer water holds less oxygen and decay is rapid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **eutrophication** → nutrient enrichment of water leading to algal growth **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	4
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** At night no photosynthesis replaces the oxygen used in respiration / decomposition so concentrations fall below the minimum fish can tolerate **1** At the end of summer large amounts of dead vegetation decay rapidly in warm water causing severe deoxygenation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deoxygenation** → loss of dissolved oxygen from the water	2
04.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Cyanobacteria release toxins harmful to livestock, pets and people **1** Floating algae shade submerged / rooted macrophytes which die, disrupting food webs **1** Algae are an unreliable food source (die off rapidly) so food webs are poor **1** Blocked filters / unpleasant taste and smell in water supplies **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 266-268; AQA spec 3.4.3.2.9.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	1 mark for the point below. **1** The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time	1
05.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Light:** UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions **1 Temperature:** higher temperatures increase the rate of chemical breakdown / microbial decomposition **1 Soil microbes / oxygen:** aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils **1** pH / presence of other chemicals may increase or decrease the rate of degradation **1** Adsorption onto clay / organic matter may protect from degradation and increase persistence **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **Adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** Most energy comes from non-renewable fossil fuels whose reserves are depleting – the rate of new discoveries has fallen sharply since the 1980s **1 Use causes unacceptable environmental damage:** atmospheric pollution, climate change, oil pollution, habitat loss **1 Economic / social:** depleting resources become more expensive, reducing future affluence – risk of an energy gap if new technologies are not developed in time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	3

Do not write outside the box

Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 4 points below. **1 Coal mining:** habitat loss (especially open-cast), spoil heaps, subsidence from deep mines, acid drainage, dust, noise **1 Oil and gas extraction:** spills from rigs, oil-based drilling mud polluting groundwater and sea, gas flaring releasing SO2 / CO2 / smoke **1** Pipelines causing habitat damage during construction – tanker transport risks **1** Fuel processing (refineries, coal washing) pollution – embodied energy of equipment and site development for all resources **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product	4
06.3	**1 mark each** 1 mark for each of the 3 points below. **1** New technologies take many years to develop, prove and scale up, and need new infrastructure (grids, storage, vehicles) **1** Existing cheap fossil fuel technologies make it hard for alternatives to compete until fuel prices rise **1** By then depletion would reduce supply faster than replacements could be built, causing shortages **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 180-186; AQA spec 3.3.3.	
07.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 25 + 5 = 30% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • A bare wrong answer with no working. • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	1

Do not write outside the box

Question	Marking guidance	Marks
07.2	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 UV:** short wavelength – mostly absorbed in the stratosphere by monatomic, diatomic and triatomic oxygen (ozone layer) which are formed and destroyed in a dynamic equilibrium – converted to chemical energy in these reactions – little reaches the surface **2 Visible light:** transmitted through the atmosphere easily (some reflected by clouds / dust) – absorbed by the Earth's surface and converted to heat – some absorbed by chlorophyll and converted to chemical energy in photosynthesis **2 IR:** emitted by the warm surface (long-wave) – absorbed by greenhouse gases (CO2, water vapour, methane) and converted to heat – the warm atmosphere re-emits IR to the surface and reduces conduction losses – some escapes to space **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV • **long-wave** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	6
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Greenhouse gases absorb most of the IR emitted by the surface and re-radiate it in all directions, including back down **1** The surface is repeatedly warmed and re-emits, so the total IR exchanged between surface and atmosphere exceeds the incoming solar energy – only the net loss (70%) balances the absorbed solar energy at equilibrium **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
07.4	1 mark for the point below. **1** The atmosphere forms layers (troposphere, stratosphere, mesosphere) with different temperature gradients because energy is absorbed at different altitudes (eg UV heating the stratosphere, surface heating the troposphere) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87-89; AQA spec 3.2.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Volume of ash is much less than the original waste (less landfill) **1** Heat used for district heating / electricity generation (energy recovery) **1** No sorting / complicated management needed **1** Destroys pathogens / flammable organic wastes converted to CO2 and water **1** Refuse-derived fuel can replace fossil fuels (eg cement kilns) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
08.2	**1 mark each** 1 mark for each of the 4 points below. **1** High temperature and rapid cooling reduce the formation of toxic dioxins (from organic wastes reacting with chlorine) **1** Scrubber dissolves hydrogen chloride (from chlorine in PVC) which is then neutralised with alkali / limestone **OR** removes heavy metals for storage at high pH **1** Wet waste needs extra fuel to burn and lowers combustion temperature – composting recovers its nutrient value **1** Recycling conserves the resource value of metals, glass and plastics that would otherwise be lost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Ash may contain concentrated heavy metals / toxic residues **1** Must be stored in solid form at high pH / hazardous waste landfill so the metals remain insoluble **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2

Do not write outside the box

Question	Marking guidance	Marks
08.4	**Any 1 of 2** 1 mark each for any 1 of the 2 points below, in any order, maximum 1. **1** Fuel (natural gas) needed to burn wet / non-flammable wastes **1** Cost of treating effluent gases (SO2, NOx, HCl, smoke) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 283-284; AQA spec 3.4.3.2.12.	
09.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (150 - 125) / 150 x 100 = 16.7% (accept 17%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** A fixed fee gives no incentive to save water because the bill is the same however much is used **1** Metering matches the charge to the volume used so households save money by using less (fewer hosepipes, low-flow appliances) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1 Low-water appliances:** dual-flush toilets, low-flow shower heads, efficient washing machines / dishwashers, tap timers **1 Greywater re-use:** bath / shower water used to flush toilets or water gardens **1** Rainwater collection / water butts for garden use **1 Xeriscaping:** planting drought-adapted species that need no watering **1** Repairing leaks / dripping taps **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** Old infrastructure / pipes in poor condition leak **1** Repair and good maintenance / pipe replacement / leak detection reduce wastage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.5	**1 mark each** 1 mark for each of the 2 points below. **1** Water is delivered directly to the plant roots / soil around each plant **1** Less evaporation loss from spray droplets and from wetting the whole soil surface – less runoff **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 124-126; AQA spec 3.2.2.5.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **State of matter:** solids deposit near the source, gases disperse widely in the atmosphere – density (lead dust, hydrogen cyanide at Bhopal) controls how far particles/gases travel • **Persistence / half-life:** persistent pollutants (organochlorines, CFCs, heavy metals, radioactive wastes with long half-lives) travel further, accumulate and need containment rather than dilution – non-persistent pollutants (sewage, organophosphates) can be allowed to degrade • Toxicity, specificity, mode of action (enzyme inhibition, neurotoxins) determine acceptable concentrations and which species are at risk – LD50 / MDAF / MDNF used to set doses • Solubility in water (nitrates disperse and leach - control by nitrate vulnerable zones) vs liposolubility (DDT, mercury bioaccumulate and biomagnify - control by bans / restrictions) • **Reactivity:** reactive pollutants may degrade quickly or form secondary pollutants (ozone, acid rain, PANs) - controlling primary pollutants (catalytic converters, FGD) prevents secondary pollutants • Adsorption onto clay / sediments immobilises metals and caesium – can be used deliberately (activated carbon filters, polymer adsorption) • Mutagenic / carcinogenic / teratogenic action means no safe dose – exposure kept As Low As Reasonably Achievable (ALARA) • Synergism (ozone + SO2; smoke + SO2) means pollutants must be considered together • Environmental features (temperature inversions, currents, pH, oxygen) interact with properties: Critical Pathway Analysis uses properties + environment to predict where pollutants go and to focus monitoring • **Evaluation:** properties of new chemicals can be predicted from similar ones (precautionary principle)	9

Do not write outside the box

Question	Marking guidance	Marks
	– understanding allows prevention rather than clean-up **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **Persistence** → does not break down / stays in the environment for a long time • **half-life** → the time for the activity, or the amount, to fall by half • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	
	Model answer - Level 3 (9 marks)	
	A pollutant's properties largely determine where it goes and how much harm it does, so they can be used both to predict its behaviour and to choose a control. State of matter and density govern dispersal: solids deposit near the source, gases spread widely, and dense gases such as hydrogen cyanide at Bhopal hug the ground. Persistence decides whether dilution is enough: non-persistent wastes like sewage can be discharged where they break down, but organochlorines, CFCs, heavy metals and long-lived radioisotopes travel far and accumulate, so they need containment, bans or restrictions instead. Toxicity, specificity and mode of action, quantified by LD50 and used to set acceptable doses, identify which species are at risk. Solubility explains why nitrate leaches and is controlled by nitrate vulnerable zones, while liposolubility explains why DDT and methyl mercury bioaccumulate and biomagnify along food chains. Reactive pollutants may degrade quickly or form secondary pollutants - ozone, acid rain, PANs - so controlling the primary emissions with catalytic converters or flue-gas desulfurisation prevents the secondary ones. Adsorption on clay immobilises metals and caesium and is exploited deliberately in activated-carbon and polymer filters. Mutagens and carcinogens have no safe dose, so exposure is kept As Low As Reasonably Achievable, and synergism between ozone and sulfur dioxide, or smoke and sulfur dioxide, means pollutants must be assessed together.	
	Properties interact with the environment - inversions, currents, pH, oxygen - which is why Critical Pathway Analysis combines both to predict where pollutants concentrate and to focus monitoring. The greatest value of this understanding is preventive: the properties of a new chemical can be predicted from similar ones under the precautionary principle, allowing control before release rather than clean-up afterwards.	
	Weaker answer - Level 1 (2 marks)	
	Pollutants have properties like toxicity, persistence and solubility. Toxic pollutants are dangerous. Persistent pollutants like DDT last a long time in the environment and build up in food chains. Soluble pollutants dissolve in water and get into rivers. Knowing these properties helps scientists to predict what pollutants will do and to control them, for example by banning DDT.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 1 defines three properties in isolation; Level 3 pairs every property with a prediction *and* a control strategy, which is what the question asks.	
	• Level 3 uses a wide range of accurate examples (Bhopal, nitrate zones, methyl mercury, PANs, caesium adsorption); Level 1 uses DDT twice.	
	• The final paragraph links properties to environmental conditions (CPA) and to the precautionary principle - synoptic evaluation that distinguishes the top band.	
10.2	1 mark for the point below. **1** The increase in the amount / concentration of a substance within an (individual) organism over time, because intake exceeds excretion or breakdown **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **bioaccumulation** → a pollutant building up inside one organism over time **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Scale of the problem:** how energy is used (transport, buildings, industry, water) – links between affluence and per capita use • **Transport:** bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles • **Buildings:** orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour • **Industry:** heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management • **Limits:** rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour • Evaluation with quantitative examples – comparison with the cost of new supply	25
	– conclusion that conservation is the cheapest first step but cannot alone meet demand growth *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Context:** uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • **New sources:** rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • **Management:** artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • **Conservation:** low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering – pollution control to protect sources • **Treatment:** matching quality to use – screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	

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